香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

ECO6102 / 2026–27 Term 1 (Fall)

# Welcome to ECO6102: Advanced Econometrics

Enjie (Jack) Ma

Lecture 1 · Monday 7 September 2026

# Teaching Team

## Instructor

- **Instructor**: Enjie (Jack) Ma 马恩杰 (maenjie@cuhk.edu.cn)
  - Econ Ph.D. @Cornell, Stats+Econ B.S. @UCLA
  - Assistant Professor, CUHK SZ SME
  - Specialty: Economics of Climate Change, Innovation, and Technology
  - **Lectures**: Xili Campus 6E506, Monday 09:00–11:50
  - **Office Hours**: Xili Campus 6A204, Monday 13:00–14:00(no OH today)

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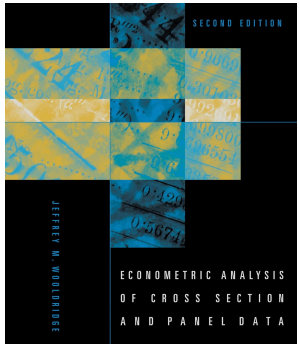
# Teaching Team

Teaching assistant and tutorial

- **Teaching Assistant**
  - Xupeng Ruan (ruanxupeng@cuhk.edu.cn)
  - **Tutorials**: Xili Campus 6E506, Monday 18:00–18:50
  - **Office Hours**: Xili Campus 6B320, Friday 09:00–10:00

# Required Textbook

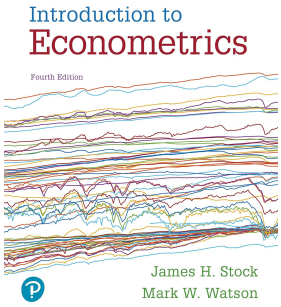
Wooldridge, *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., MIT Press, 2010




Second Edition, 2010

# Other References

Stock and Watson, *Introduction to Econometrics*, 4th ed., Pearson, 2019




Recommended for reading

# Software Package

- Data analysis is essential for learning econometrics. Be prepared to write code for analyzing real-world data.
- In this course, we will use a software package called Python. It is free and open-source. Python will be mostly taught in tutorials by the TA. You should use Python to finish your homework.
- Python has become popular among economists, so it is beneficial for you to learn this software package.

# Course Components

- Lectures: Lectures' content will center around econometric concepts based on the main textbook. In addition to the lecture slides, you should read the main textbook. (10-min break time around half way 10:20am)
- Tutorials: Tutorial sessions will be hosted by the TA. During these sessions, the TA will teach Python, discuss problem sets, explain difficult materials, and answer questions.
- Office Hours: You may stop by during my office hours with any questions or by appointment. Emails are not preferred since it is hard to type equations in emails.

# Course Website

- **Course Website**: bb.cuhk.edu.cn
- The site contains the syllabus, lecture slides, problem sets, occasional announcements, and other materials.
- You should check it regularly as you are responsible for any information posted there in addition to any announcements made in class.
- Throughout the semester, I will make announcements via the course website.

# Grading Policy

- **Participation (10%)**
- **Problem Sets (20%)**: Group study and free discussion are encouraged, but you should submit your own answers. No late assignments will be accepted.
- **Midterm Exam (30%) and Final Exam (40%)**: There will be no make-up exams. In case of illness or accident at the time of the midterm—with proper documentation from a doctor or the police—the final will be weighted 70%.
- **Regrading Request**: If you think a mistake was made in grading your exam, you may ask for a regrade. See the syllabus for more information.

# Learning Goals

By the end of the course, you should:

- Be skilled in using econometric tools and capable of critically evaluating quantitative analyses conducted by other researchers.
- Be able to select appropriate empirical methods to tackle practical questions in economics and related fields, while understanding the assumptions, limitations, and potential improvements of these methods.
- Be able to use the statistical software package Python for empirical analysis and simulation studies.

# Course Outline

- Introduction (Chapter 1)
- Conditional Expectations (Chapter 2)
- Linear Regression (Chapter 4)
- Instrumental Variables (Chapter 5)
- Simultaneous Equation (Chapter 9)
- Panel Data (Chapters 10 & 11)
- Binary Response (Chapter 15)

# Any Questions?



Lecture 1:

# Introduction

# Roadmap

L1 | L2 | L3 | L4 | L5 | Midterm | L6 | L7 | L8 | Final

Introduction — Wooldridge (2010), Chapter 1 (Week 1)

1 Causality and Endogeneity

2 Data Structures

3 Examples and Random Regressors

SECTION 1

# Causality and Endogeneity

# Causal Inference

The goal of most empirical studies in economics and other social sciences is to determine whether a change in one variable $w$ causes a change in another variable $y$. For example,

- Does having another year of education cause an increase in monthly salary?
- Does reducing class size cause an improvement in student performance?
- Does lowering the business property tax rate cause an increase in the city's economic activity?
- Does access to AI coding tools raise a programmer's pay?
- Other examples?

# Correlation versus Causality

## Correlation and causal relationships

- Because economic variables are properly interpreted as random variables, we should use ideas from probability to formalize the sense in which a change in $w$ causes a change in $y$.
- We learned from probability how to determine whether two random variables are correlated (How?).

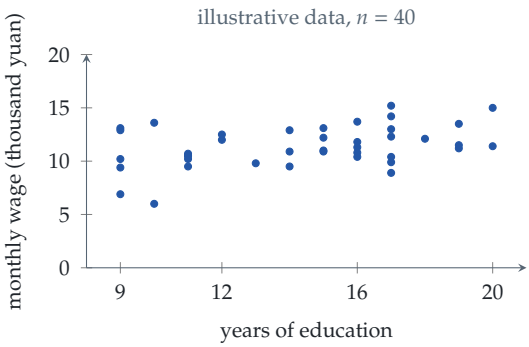
# Correlation versus Causality

## Detecting correlation



# Correlation versus Causality

## Detecting correlation




$$\rho_{wy} = \frac{\mathrm{Cov}(w, y)}{\sigma_w \, \sigma_y} \in [-1, 1]$$

- Here $w$ is education and $y$ is monthly wage.
- **Sample analogue**: $\hat{\rho}$ from the $n$ pairs $(w_i, y_i)$.
- $\rho$ measures co-movement; it is silent about which causes which.

# Correlation versus Causality

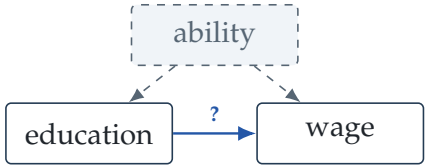
## Education and earnings

- But correlation $\neq$ causality!

# Correlation versus Causality

## Education and earnings

- But correlation ≠ causality!
- For example, individuals with higher levels of education tend to earn more (correlation).
- However, those with more education are also likely to have higher abilities.
- We do not know whether the increase in salary is caused by education or ability.



# Ceteris Paribus Analysis

## Holding other factors fixed

- The notion of **ceteris paribus**—that is, holding all other factors fixed—is at the crux of establishing a causal relationship.
- Simply finding that two variables are correlated is rarely enough to conclude that a change in one variable causes a change in another.
- This result is due to the nature of economic data: rarely can we run a controlled experiment that allows a simple correlation analysis to uncover causality.

# Ceteris Paribus Analysis

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- The notion of **ceteris paribus**—that is, holding all other factors fixed—is at the crux of establishing a causal relationship.
- Simply finding that two variables are correlated is rarely enough to conclude that a change in one variable causes a change in another.
- This result is due to the nature of economic data: rarely can we run a controlled experiment that allows a simple correlation analysis to uncover causality.
- Instead, we can use econometric methods to effectively hold other factors fixed – core of this class, using econometric model to establish causality.

**The core of this course: using econometric models to establish causality.**

# Ceteris Paribus Analysis

## The average response

- Since economic variables are random, we cannot predict the response of each individual; instead, we focus on the average response.
- A ceteris paribus analysis entails estimating $E(y|w, \mathbf{c})$, the expected value of $y$ conditional on $w$ and $\mathbf{c}$.
- The vector $\mathbf{c}$ denotes a set of control variables that we would like to explicitly hold fixed when studying the effect of $w$ on the expected value of $y$.
- The reason we control for these variables is that we think $w$ is correlated with other factors that also influence $y$.

# Partial Effect

- If $w$ is continuous, interest centers on $\partial E\left(y|w,\mathbf{c}\right)/\partial w$, which is usually called the partial effect of $w$ on $E\left(y|w,\mathbf{c}\right)$.
- If $w$ is discrete, we are interested in $E\left(y|w,\mathbf{c}\right)$ evaluated at different values of $w$, with the elements of $\mathbf{c}$ fixed at the same specified values.

$$E(wage \mid educ+1,\mathbf{c}) - E(wage \mid educ,\mathbf{c})$$

- One extra year of schooling; the same controls in both terms.

# Controls

## What are "good controls" $\mathbf{c}$?

- Deciding on the list of proper controls is not always straightforward, and using different controls can lead to different conclusions about a causal relationship between $y$ and $w$.
- This is where establishing causality gets tricky: it is up to us to decide which factors need to be held fixed.
- If we settle on a list of controls, and if all elements of $\mathbf{c}$ can be observed, then estimating the partial effect of $w$ on $E(y|w, \mathbf{c})$ is relatively straightforward.

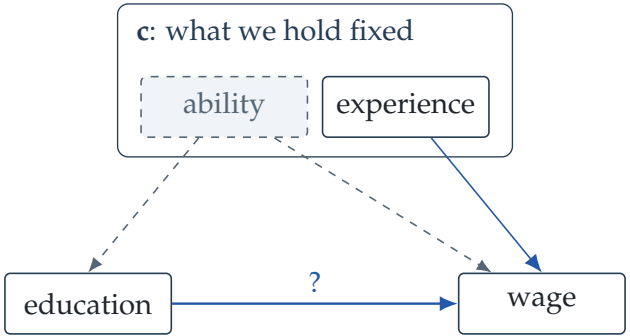
# Controls

## Can you observe them?

- Unfortunately, in economics and other social sciences, many elements of $\mathbf{c}$ are not observed.
- For example, in estimating the causal effect of education on wage, we might focus on $E(wage|educ, exper, abil)$, where $educ$ is years of schooling, $exper$ is years of workforce experience, and $abil$ is innate ability.
- In this case, $\mathbf{c} = (exper, abil)$, where $exper$ is observed but $abil$ is not.

# Controls

Can you observe them? (continued)




- Dashed box = not observed; dashed arrows = paths that run through it.

# Controls

## Attendance and exam performance

- As a second example, consider establishing a causal relationship between student attendance and performance on a final exam in a principles of *economics* class.
- We might be interested in $E(score|attend, SAT, priGPA)$, where *score* is the final exam score, *attend* is the attendance rate, *SAT* is the score on the Scholastic Aptitude Test, and *priGPA* is the grade point average at the beginning of the term.
- We can reasonably collect data on all of these variables for a large group of students.

# Controls

## Remaining unobserved factors

- Is this setup enough to decide whether attendance has a causal effect on performance?

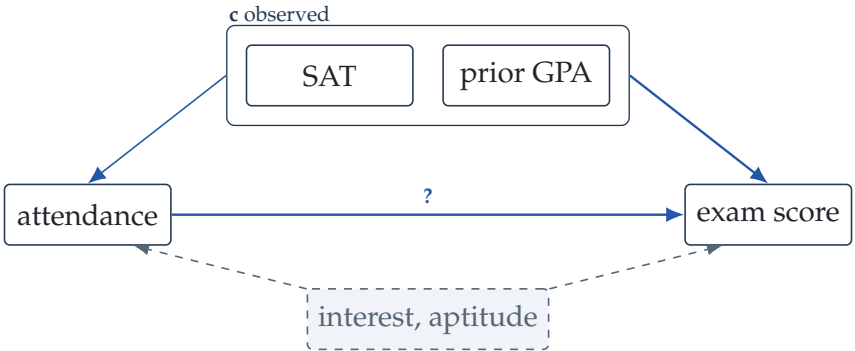
# Controls

## Remaining unobserved factors

- Is this setup enough to decide whether attendance has a causal effect on performance? Maybe not.
- While *SAT* and *priGPA* are general measures reflecting student ability and study habits, they do not necessarily measure one's interest in or aptitude for economics.
- Such attributes, which are difficult to quantify, may nevertheless belong in the list of controls if we are going to be able to infer that the attendance rate has a causal effect on performance.

# Controls

## Remaining unobserved factors (continued)




- Observed controls can be held fixed; interest and aptitude cannot.

# Controls

What makes setting good control hard?

- **You choose c**: no formula gives the list; it takes institutional knowledge of the question.
- **Not everything belongs**: choose controls for the causal question; adding variables is not automatically better.
- **Part of c is unobservable**: ability, interest, management quality — the dashed boxes.
- **The data may not exist**, and different **c** give different answers — always report which controls you used.

# What Else Can Go Wrong?

## Measurement error and simultaneity

- In addition to not being able to obtain data on all desired controls (omitted variables), other problems can interfere with estimating causal relationships.
- For example, even if we have good measures of the elements of $\mathbf{c}$, we might not have very good measures of $y$ or $w$ (measurement error).
- A more subtle problem is that we may only observe equilibrium values of $y$ and $w$ when these variables are simultaneously determined (reverse causality).
  - An example is determining the causal effect of conviction rates ($w$) on city crime rates ($y$).

# Obstacles to Establishing causality

## Section summary

- **Confounders**: correlation is not causality.
- **Controls**: you choose $\mathbf{c}$, and part of it is unobserved (omitted variables).
- **Measurement error**: classical error in $y$ adds noise; error in $w$ can bias the estimated effect.
- **Simultaneity**: we observe only equilibrium values.
- These problems can make $u$ related to the regressors: endogeneity. Whether this happens depends on the model.

SECTION 2

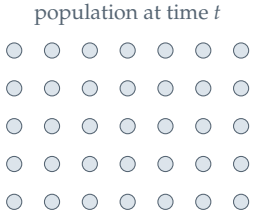
# Data Structures

# Data Structures

- For much of this course, we adopt a random sampling assumption.
- More precisely, we assume that
  1. A population model has been specified and
  2. An independent, identically distributed (i.i.d.) sample can be drawn from the population.
- Specifying a population model—which may be a model of $E(y|w, \mathbf{c})$—requires us first to clearly define the **population of interest**.

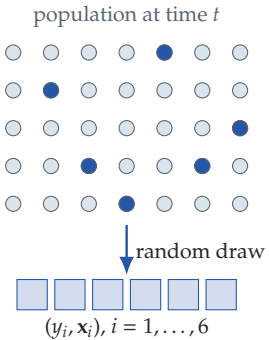
# Cross Section Data

- Because we will rely heavily on random sampling, it is important to know what it allows and what it rules out.
- Random sampling is often reasonable for cross section data, where, at a given point in time, units are selected at random from the population.
- In this setup, any explanatory variables are treated as random outcomes, along with data on response variables.



# Cross Section Data

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# Pooled Cross Section Data

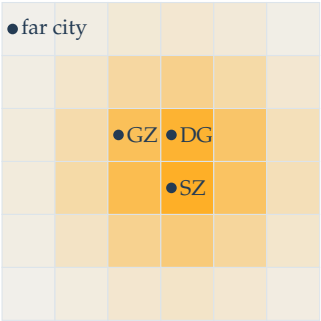
- Random sampling does exclude cases of some interest for cross section analysis.
- For example, the identical distribution assumption is unlikely to hold for a pooled cross section, where random samples are obtained from the population at different points in time.
- Example: a random sample of workers in 2019 and another in 2020, stacked. Each year is a fine random sample, but a 2020 worker is not drawn from the same population as a 2010 worker.
- This case is covered by independent, not identically distributed (i.n.i.d.) observations.

# Spatial Data

- A situation that does require special consideration occurs when cross section observations are not independent of one another.
- An example is spatial correlation models. This situation arises when dealing with large geographical units that cannot be assumed to be independent draws from a large population, such as the 50 states in the United States.
- It is reasonable to expect that the unemployment rate in one state is correlated with the unemployment rate in neighboring states.
- Spatially correlated shocks are everywhere in *environmental economics*: a heat wave, a flood or a smog episode hits neighbouring firms and cities together.

# Spatial Data

## Neighbours share shocks




temperature anomaly, one week

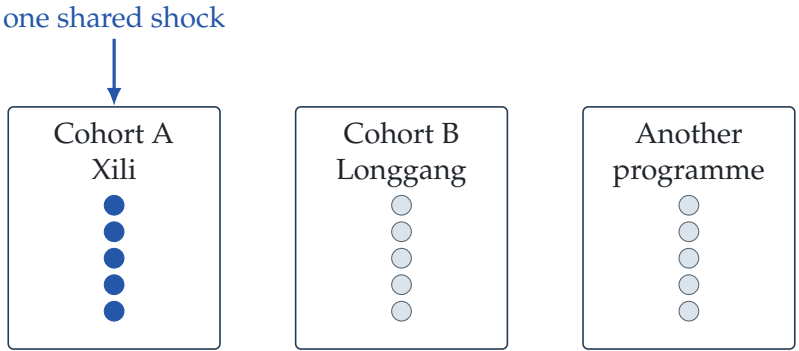
- A heat wave ignores city borders: Shenzhen, Dongguan and Guangzhou are hot in the same week.
- Firms next door share the shock. If it is not in the model it sits in $u$, so their errors are correlated: the estimate may be fine, the standard errors are not.

# Cluster Data

- Cluster sampling also induces correlation in a cross section data set, but in most cases, it is relatively easy to deal with econometrically.
- For example, retirement savings of employees within a firm may be correlated because of common characteristics of workers within a firm or because of features of the firm itself (e.g., retirement plan).
- Each firm represents a group or cluster, and we may sample several workers from a large number of firms.
- Provided the number of clusters is large relative to the cluster sizes, standard methods can correct for the presence of within-cluster correlation.

# Cluster Data

## Groups share shocks

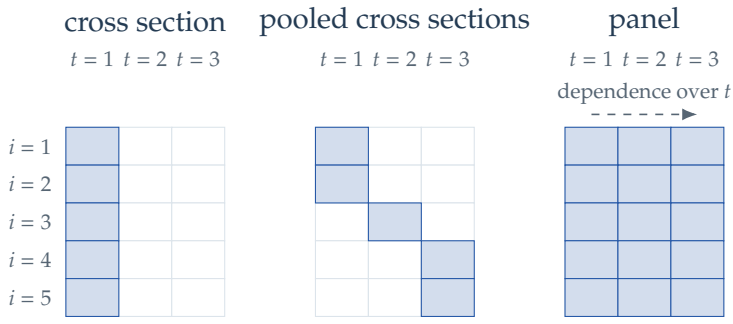


# Panel Data

- For panel data (or longitudinal data), which consists of repeated observations on the same unit (e.g., individuals, households, firms, cities) over time, the random sampling assumption initially appears much too restrictive.
- However, the random sampling assumption, properly stated, does allow for temporal correlation.
- What we will do is to assume random sampling in the cross section dimension. The dependence in the time series dimension can be entirely unrestricted.
- This approach is justified in panel data applications with many cross section observations spanning a relatively short time period.

# Cross Section versus Panel

## One grid




- *i* runs down the rows (units), *t* across the columns (time); a panel keeps the same rows in every period: random across rows, anything goes along a row.

# Why Data Structure Matters

- **Structure tells you who may share shocks**: separate units (cross section), the same unit over time (panel), neighbours (spatial), members of a group (cluster).
- **Shared shocks change inference**: we must justify independence across sampled units or account for their dependence. A cross section is not automatically i.i.d.
- **It does not remove confounding**: the controls problem of Section 1 is still there. Structure is the second question, not a substitute for the first.

SECTION 3

# Examples and Random Regressors

# Some Examples

## Example 1.1: Wage Offer Function

- Earlier we used wage in money units. This model uses $\log(wage^o)$: the natural log of the wage offer.

$$\log(wage^o) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 married + u$$

- *educ*: years of schooling; *exper*: years of work experience; *married*: a binary indicator. The superscript $o$ means offer.
- The error term $u$ contains unobserved wage determinants; the $\beta_j$ are the unknown parameters.

# Some Examples

## Example 1.1: Wage Offer Function

$$\log(wage^o) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 married + u$$

- **Population: working women**. We observe wages for the women in this population.
- **Population: all women over age 18**. A random sample also includes women who are not working; their wage offers are unobserved.
- Randomly sampling people does not ensure that the outcome is observed for everyone.

# Some Examples

## Example 1.1: Wage Offer Function

$$\log(wage^o) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 married + u$$

- Write $\mathbf{x} = (educ, exper, married)$. The key assumptions concern $u$ given $\mathbf{x}$ in the population.

**1. Conditional mean**

$$E(u \mid \mathbf{x}) = 0\,?$$

Do observed characteristics predict unobserved wage determinants on average?

**2. Conditional variance**

$$\mathrm{Var}(u \mid \mathbf{x}) = \sigma^2\,?$$

Is the spread of unobserved wage determinants the same across groups?

# Some Examples

## Example 1.1: Wage Offer Function

- For cross section data, it is useful to write the model as a generic draw from the population

$$\log\left(wage_i^o\right) = \beta_0 + \beta_1 educ_i + \beta_2 exper_i + \beta_3 married_i + u_i$$

  where $i$ indexes individuals.
- Throughout the course, the $i$ subscript is reserved for indexing cross section units (e.g., individuals, firms, cities).
- The $t$ subscript is reserved for time (next example); $j$, $g$, $h$ index variables, parameters and equations.

# Some Examples

## Example 1.2: Effect of Spillovers on Firm Output

- Suppose that the population is all manufacturing firms in a country operating during a given three-year period.
- A production function describing output in the population of firms is

$$\begin{aligned} \log(output_t) &= \delta_t + \beta_1 \log(labor_t) + \beta_2 \log(capital_t) \\ &\quad + \beta_3 spillover_t + quality + u_t, \quad t = 1, 2, 3. \end{aligned}$$

- Here, $spillover_t$ is a measure of foreign firm concentration in the region containing the firm.

# Some Examples

## Example 1.2: Effect of Spillovers on Firm Output

$$\begin{aligned} \log(output_t) &= \delta_t + \beta_1 \log(labor_t) + \beta_2 \log(capital_t) \\ &\quad + \beta_3 spillover_t + quality + u_t, \quad t = 1, 2, 3. \end{aligned}$$

- The term *quality* contains unobserved factors (e.g., unobserved managerial, worker quality), which affect productivity and are constant over time.
- The error $u_t$ represents unobserved shocks in each time period.

# Some Examples

## Example 1.2: Effect of Spillovers on Firm Output

$$\begin{aligned} \log(output_t) &= \delta_t + \beta_1 \log(labor_t) + \beta_2 \log(capital_t) \\ &\quad + \beta_3 spillover_t + quality + u_t, \quad t = 1, 2, 3. \end{aligned}$$

- When we study panel data methods, there are several issues in deciding how best to estimate the $\beta_j$.
- An important one is whether the unobserved productivity factors *quality* are correlated with the observable inputs $labor_t$ and $capital_t$.
- Also, can we assume that $spillover_t$ at, say, $t = 3$ is uncorrelated with the error terms in all time periods?

# Some Examples

## Example 1.2: Effect of Spillovers on Firm Output

- For panel data, it is especially useful to add an $i$ subscript indicating a generic cross section observation:

$$\begin{aligned}\log(output_{it}) &= \delta_t + \beta_1 \log(labor_{it}) + \beta_2 \log(capital_{it}) \\ &\quad + \beta_3 spillover_{it} + quality_i + u_{it}, \quad t = 1, 2, 3.\end{aligned}$$

- $quality_i$ is a firm-specific term that is constant over time and also has the same effect in each time period, while $u_{it}$ changes across time and firm.
- The assumptions describe a generic firm $i$ drawn from the population.

# Some Examples

## Example 1.2: Effect of Spillovers on Firm Output

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- The assumptions describe a generic firm $i$ drawn from the population.

**Bonus: what could $\delta_t$ capture for a firm?**

# Fixed Regressors?

- In the above examples, the error in an equation can be correlated with one or more of the explanatory variables.
- This possibility is common in social science applications.
- The assumption of fixed regressors rules out such correlation a priori.

# Shortcoming of Fixed Regressors

## The fixed-regressor assumption

- To illustrate the shortcomings of the fixed regressor assumption, consider a linear model for cross section data:

$$y_i = \beta_0 + \mathbf{x}_i\boldsymbol{\beta} + u_i, \quad i = 1, 2, \ldots, N.$$

- In the wage example: $y_i = \log(wage_i^o)$ and $\mathbf{x}_i = (educ_i, exper_i, married_i)$. In general, $\mathbf{x}_i$ is $1 \times K$ and $\boldsymbol{\beta}$ is $K \times 1$.
- The "ideal" assumptions for this model are commonly stated as "The errors $\{u_i, i = 1, 2, \ldots, N\}$ are i.i.d. with $E(u_i) = 0$ and $E(u_i^2) = \sigma^2$."
- This statement omits the most important consideration: What is assumed about the relationship between $u_i$ and $\mathbf{x}_i$?

# Shortcoming of Fixed Regressors

Implications for the error term

$$y_i = \beta_0 + \mathbf{x}_i\boldsymbol{\beta} + u_i$$

- If the $\mathbf{x}_i$ are taken as nonrandom (i.e. constant), then $u_i$ and $\mathbf{x}_i$ are independent of one another.
- In nonexperimental environments, this assumption rules out too many situations of interest.
- Some important questions, such as efficiency comparisons across models with different explanatory variables, cannot even be asked in the context of fixed regressors.
- Random $\mathbf{x}_i$ forces the real questions: $E(u \mid \mathbf{x}) = 0$? $\text{Var}(u \mid \mathbf{x})$ constant?
- Thus we assume random regressors throughout the course.

# Wrap-up

# Key Takeaways

- Ceteris paribus: holding all other factors fixed; correlation ≠ causality
- Partial effects and control variables, observed and unobserved
- Endogeneity: $u$ related to the regressors — omitted variables, mismeasured regressors, simultaneity
- Data structures: cross section, pooled cross section, spatial, cluster, panel
- Random regressors, not fixed, throughout the course

# Next Lecture

- Today: Introduction
  Wooldridge (2010), Chapter 1
- Next: Conditional Expectations
  Wooldridge (2010), Chapter 2 (Week 2)

Adapted from slides by Shuyang Sheng and Qihui Chen

# Attendance and Background survey

https://v.wjx.cn/vm/h0uQynL.aspx#

